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Starboard Maritime Intelligence

National Maritime Security and Defence

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starboardintelligence.com

Multi-source data fusion

3B+ data points analysed daily

1 AIS DATA (+ VMS WHERE SUPPLIED)
2 ENVIRONMENTAL DATAFOR CONTEXT
3 VESSEL DATAINCLUDING OWNERSHIP
4 CUSTOMER TAGS AND WATCHLISTS
5 SATELLITE + ASSET DATA
6 NATO DATA FORMATS (i.e STANAG 6503 etc)

AI behaviour detection

300M verified vessel positions

AI BEHAVIOUR DETECTION

ANCHORAGE BEHAVIOUR

RISK ANALYSIS

FISHING BEHAVIOUR

SATELLITE VESSEL DETECTIONS

SABOTAGE SECURITY

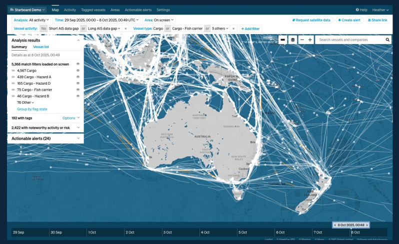
DATA CLEANING & INTEGRITY

METEOROLOGICAL ANALYSIS

Actionable, not just visible

Response teams receive alerts in **real-time**

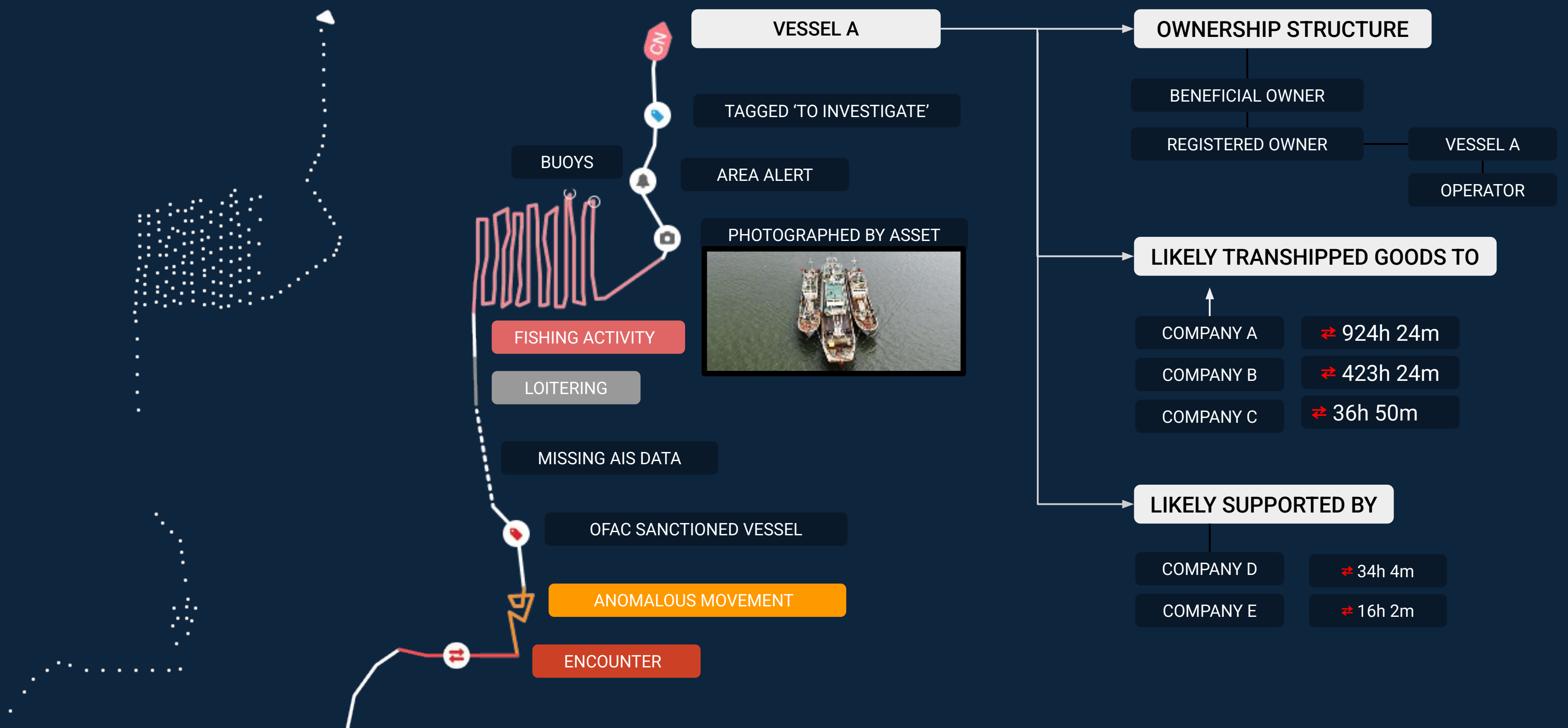
CONFIGURATION & IDENTIFICATION



SHAREABLE DATA

- UI exports as csv/kml
- URL link sharing
- Email alerts
- Analytics environment
- API push/pull

Raw AIS and other data → Behavioural insights → Vessel info & relationships



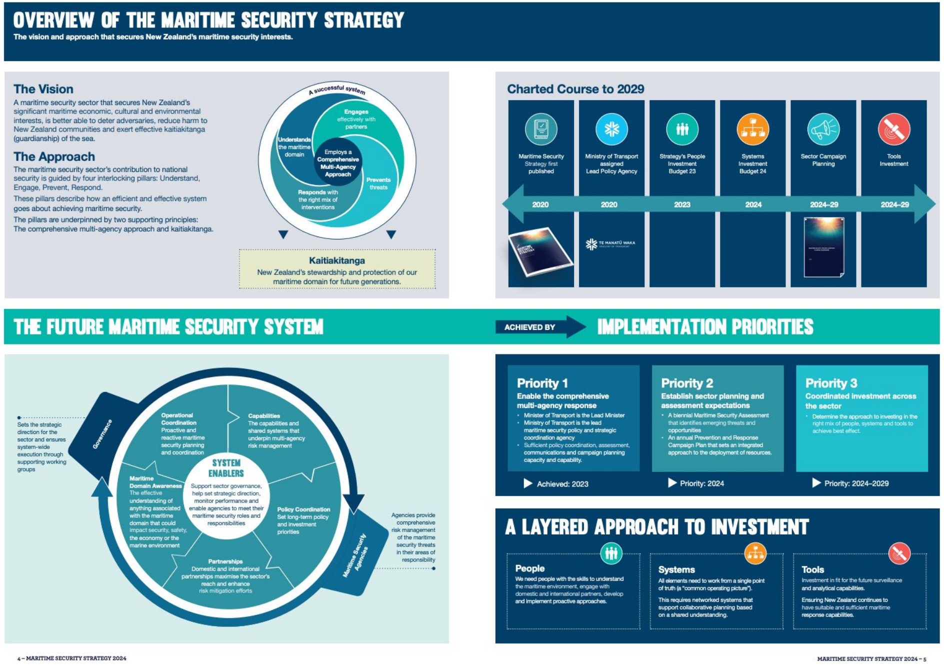
National Maritime Security

Maritime Security Strategy

The 2024 New Zealand Maritime Security Strategy called for the development of New Zealand's all of government approach to the guardianship of New Zealand's vast maritime domain.

Critical to the strategy was the system: **All elements need to work from a single point of truth (a "common operating picture").**

This requires networked systems that support collaborative planning based on a shared understanding.



National Maritime Security

Starboard Maritime Intelligence

Data Fusion System

In July 2024 the New Zealand Government, through the New Zealand Ministry of Defence, selected Starboard Maritime Intelligence.

Starboard links 14 New Zealand agencies into a unified network, allowing the seamless and protected sharing of information.

As a software as a service provider Starboard provided capability from day 1, and have continued to build additional features and capabilities to meet the needs of users.

MANATŪ KAUPAPA WAONGA
NEW ZEALAND MINISTRY OF DEFENCE

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Improving maritime awareness

12 Jul 2024

Defence has signed a contract with a New Zealand business for a service to improve New Zealand's maritime awareness.

A project for improved protection against maritime threats received $14.6 million in funding in Budget 24, and Cabinet has approved the final business case.

After an extensive international search, a contract for a subscription to a cloud-based platform is being awarded to New Zealand company Starboard Maritime Intelligence. The system will provide a real-time picture of what is happening in New Zealand's extensive maritime domain and provide a global history of arriving vessels.

"Starboard is a New Zealand technology success story. It is a spin-off from a company which received MBIE funding as a regional research institute and is now offering a service that will benefit not only Defence, but all of New Zealand," Sarah Minson, Deputy Secretary Capability Delivery at the Ministry of Defence said.

The platform will increase New Zealand's ability to detect and respond to malicious activity, natural disasters and potentially hostile vessels including illegal fishing, in our exclusive economic zone and across the Pacific. It will be used by multiple government agencies.

This project responds to the need - as outlined in the Maritime Security Strategy - for continued and improved maritime domain awareness as essential for the security and prosperity of New Zealand.

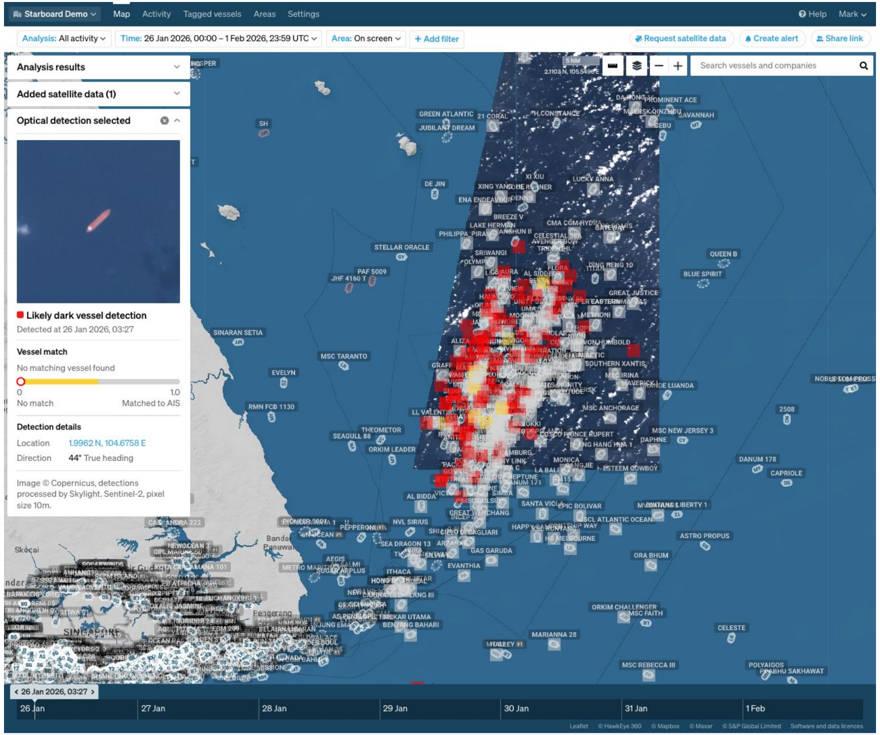
The system will make use of existing, global commercial feeds as well as specific New Zealand Government datasets in order to assess what is occurring in our maritime domain. The capability will be managed by the National Maritime Coordination Centre (NMCC) and the platform will be up and running immediately.

CAPABILITIES

Next Generation Maritime Domain Awareness

Copernicus Sentinel Optical imagery showing dark tankers and associated vessels in the Johor Anchorage.

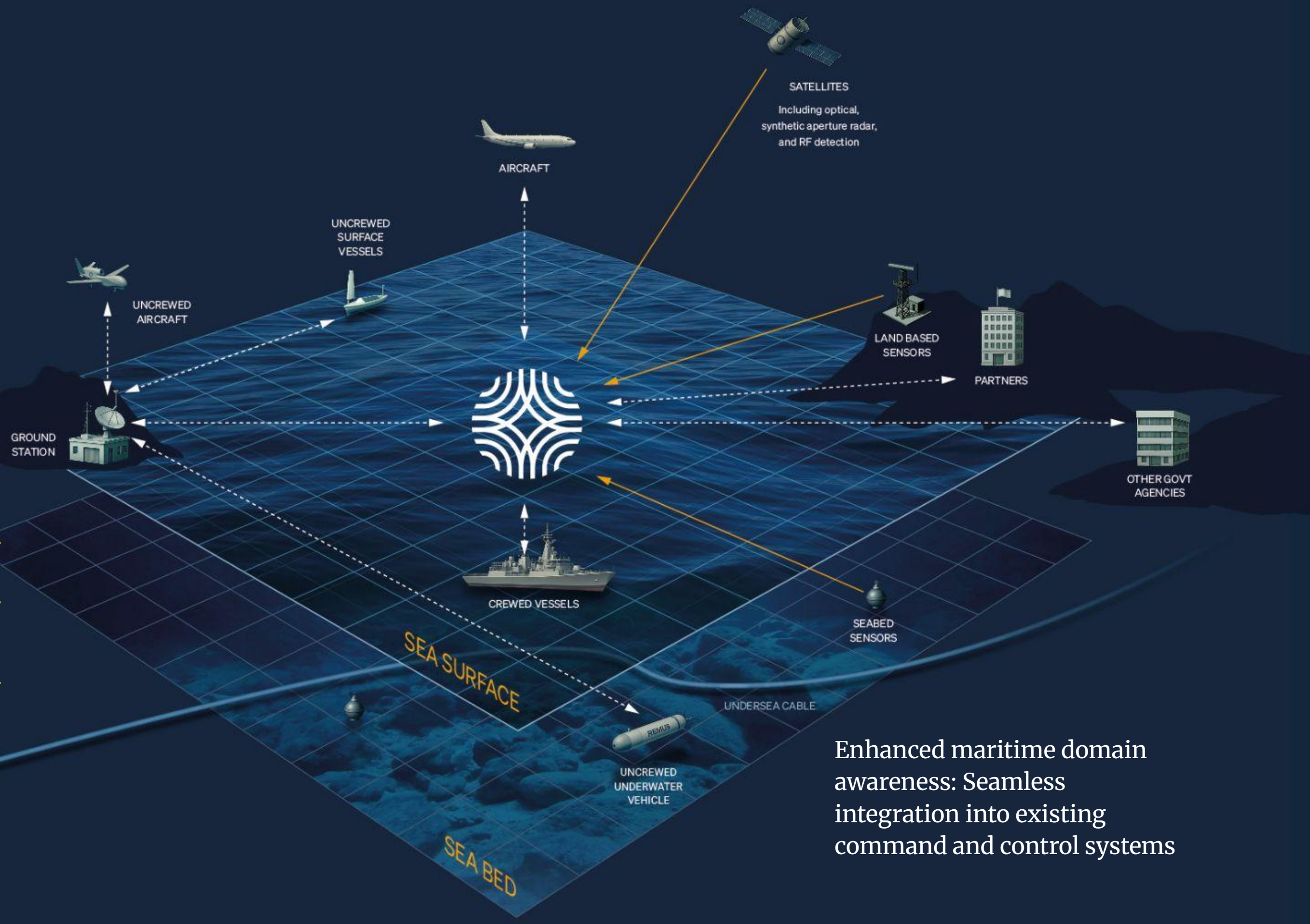
Starboard combines imagery, third party vessel detections, and propriety AIS matching to provide persistent, global detections of dark, and other high risk vessels.



The future of maritime intelligence

System agnostic

- ✓ Predictive behaviour algorithms.
- ✓ Increases common multi-domain situational awareness.
- ✓ Data fusion and integration of complementary technologies and sensors (EO/IR, Radar, LiDAR, Sonar, weapons systems, etc) in command and control systems.



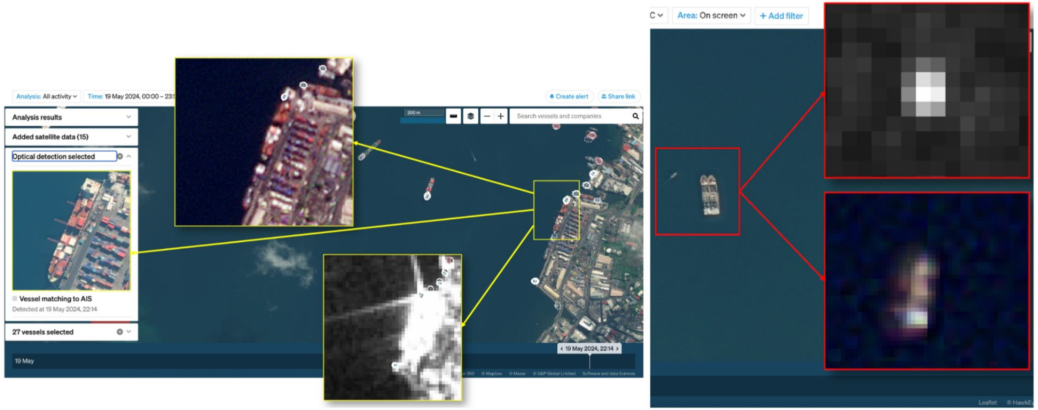
Enhanced maritime domain awareness: Seamless integration into existing command and control systems

IMAGERY

Satellite Sensing in Starboard

Starboard integrates with all major satellite imagery providers, offering a wide array of options to meet diverse operational needs. Sensor types include:

- Persistent optical and synthetic aperture radar
- High resolution cued
- Radio frequency detection
- Novel sensor types

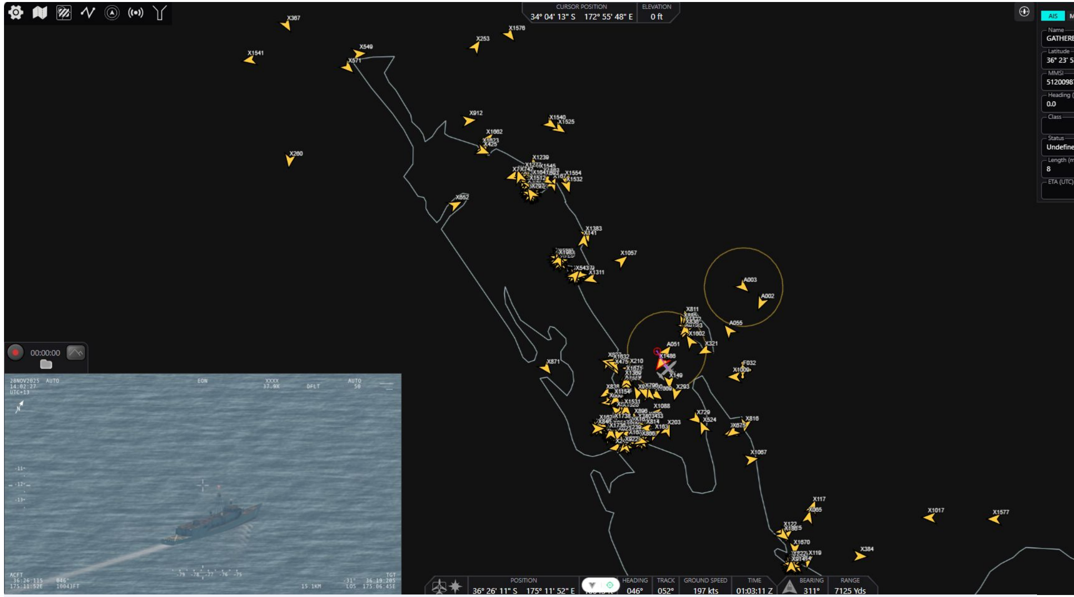


A dockside cargo vessel of 180 m length imaged in different resolutions on the same day. The background image and the insert of the left is at 0.4 m resolution (Maxar), the insert in the centre-top is at 3 m resolution (PlanetScope), and the insert at the cent bottom is a SAR image at 10 m resolution (Sentinel-1). The two side-by-side vessels are of 35 m length. The low resolution sensors detect these vessels, but provide little identifying detail. There is also a small speedboat visible in the high resolution image, which is not detected at all in the lower resolution imagery.

Capabilities

Mission System Integration

Starboard provides a range of deployment options, including edge-deployable to provide fusion and analysis where it matters most. The system can provide immediate target feedback to C2/C4ISR system, with an intuitive tip-and-cue workflow.

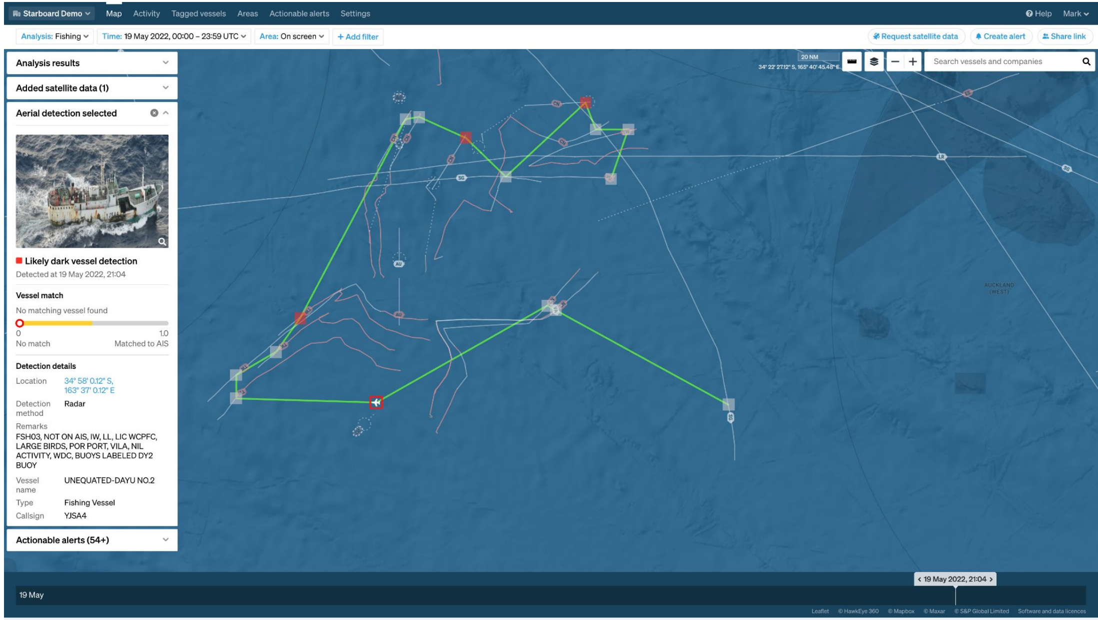


Starboard vessel tracks shown in the RNZAF C-130J mission system simulator

CAPABILITIES

Streamlining data flow from deployed patrol assets to analysts.

Starboard can receive data from deployed patrol assets, providing real time analysis capabilities to analysts ashore.



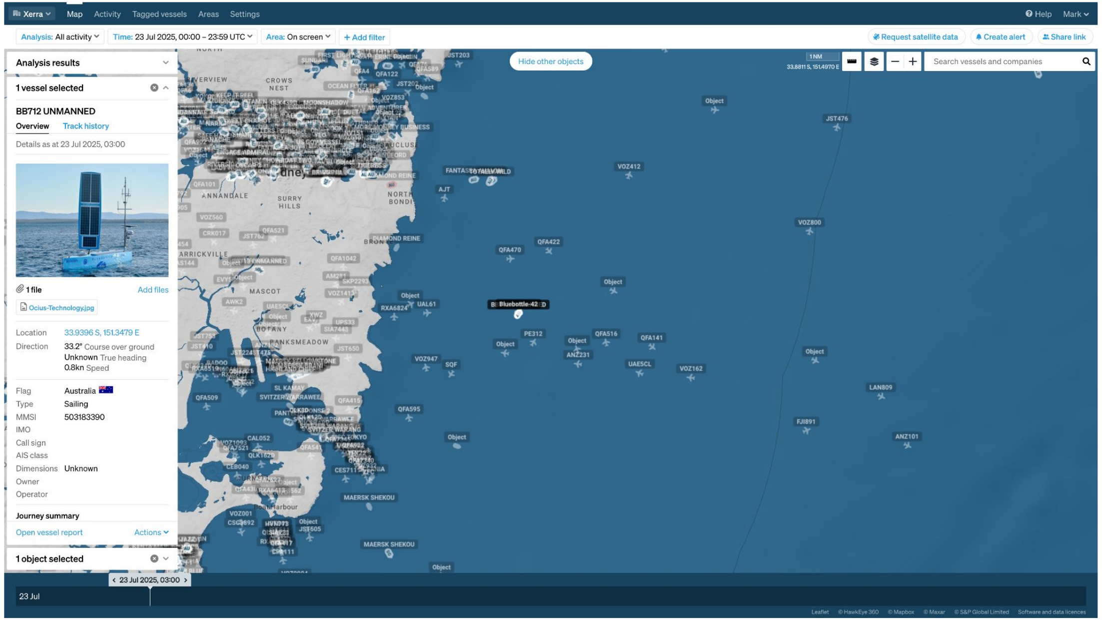
RNZAF P-3 Orion maritime patrol over the Tasman Sea

CAPABILITIES

Starboard Maritime Intelligence

Extending reach with uncrewed systems

Starboard integrates and fuses data from uncrewed systems, whether subsea, surface, or aerial.

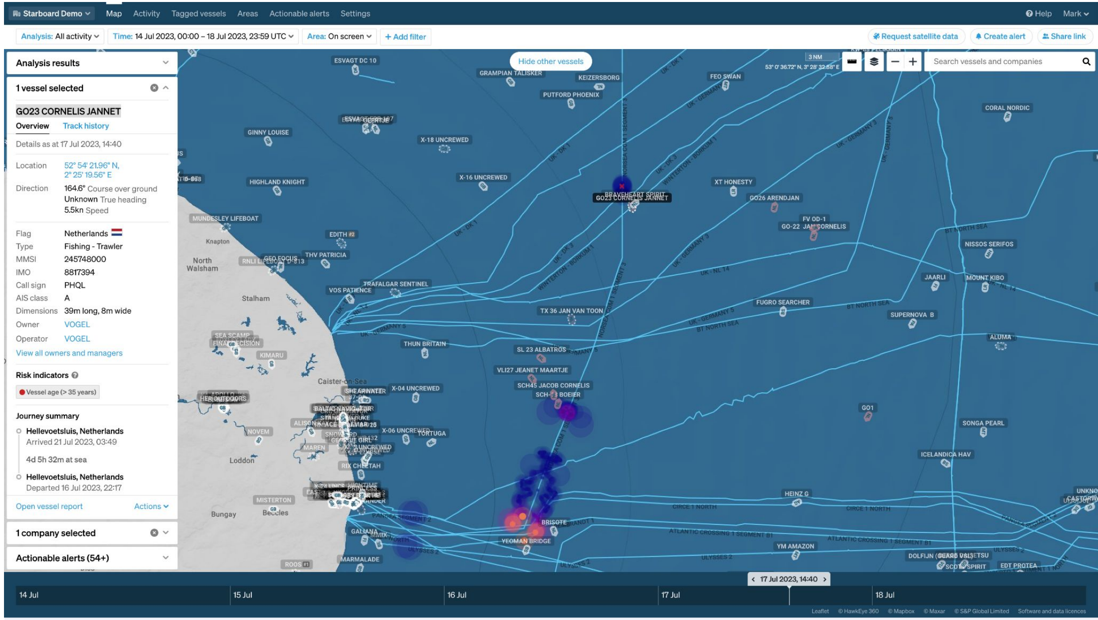


Ocius Bluebottle uncrewed surface vessel streaming surface and air contacts into Starboard for immediate use by analysts.

CAPABILITIES

From Space to Seabed

Passive acoustic sensors provide underwater detections, fused with Starboard's surface picture to zero in on unattributed sensor contacts.



Distributed acoustic sensing data from seabed cables detect the passage of vessels. This information can then be correlated with AIS

starboardintelligence.com

Thank you

Mark Douglas Maritime Domain Alayst

Starboard
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ACTIONABLE MARITIME INTELLIGENCE

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